

From a paper algorithm to Python

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Four packs contain six counters each. What steps find and show the total?

My starting explanation

2. Practise with fading help

Use W2: Predict then observe for From a paper algorithm to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for From a paper algorithm to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace variables and a change

Predict the printed value. Change only packs to 5 and predict again. Explain which statements apply a written rule.

```
packs = 3  
per_pack = 7  
total = packs * per_pack  
print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Connect the paper steps to Python

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

6. W2 Predict then observe

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual output, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

7. W3 Explain an order error

Move `print(total)` to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

8. W4 Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using `trays`, `per_tray` and `tiles`. Predict its output. Then change only `trays` to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

9. W5 Keep an honest conclusion

Explain the difference between a successful run and a correct calculation. State one limit of this program and what you would keep so another person can check your work.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled from a paper algorithm to python evidence sheet to Python readiness portfolio. Show how you can translate a short paper calculation into a python script. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: From a paper algorithm to Python

Convert a seed-count algorithm. The paper plan counts seeds in identical trays, then adds a labelled spare tray.

Python 3: `trays=4; per_tray=6; spare=2; total=trays*per_tray+spare; print(total)`. All quantities count seeds.

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From a paper algorithm to Python

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the inputs

Four trays hold six seeds each; two extra seeds are separate.



2. Translate the plan

Multiply tray count by seeds per tray, then add the spare count.



3. Trace the assignment

$4 \times 6 + 2$ is 26; total stores 26 and print displays it.

CHANGE ONE THING

Change spare to 5 while keeping both tray inputs fixed.

Explain the changed result using the matching lesson.

Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Convert a seed-count algorithm. The paper plan counts seeds in identical trays, then adds a labelled spare tray.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? $4*6+2$ is 26; total stores 26 and print displays it.

1. Trace the assignment
2. Name the inputs
3. Translate the plan

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Trace the assignment $4*6+2$ is 26; total stores 26 and print displays it. Python follows the arithmetic instructions. This translation automates a declared

counting rule; it does not train a model from examples.

Practice 2: explain the deciding evidence

Changed-case practice: Change spare to 5 while keeping both tray inputs fixed. Which conclusion is supported by the supplied facts?

1. The printed total becomes 29.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The printed total becomes 29. 4×6 remains 24; adding five gives 29.

Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
